Practical Experiment: Compare the response speed of open-source Large Language Models (LLMs) to find the fastest model for real-time applications

Meeting 3 Data Extraction & DataFrame Export and Import

Presented by: Mariyan Vasilev Apostolov
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Agenda

- Importance of Data Export
- DataFrame – The Heart of Data Processing
- Data CSV Export Flow
- Data CSV Import Flow
- Introduction to Google Cloud Platform (GCP)
- Google BigQuery - Cloud Data Warehouse
- Identity and Access Management (IAM) & Service Account Authentication
- BigQuery Export Flow
- BigQuery Import Flow
- CSV vs BigQuery Flow Comparison

Importance of Data Export

Core Reasons:

Data Persistence & Backup
- Evaluation results are valuable intellectual resource
- Local CSV files serve as backups
- Cloud storage ensures disaster recovery
- Prevents data loss from runtime crashes

Collaboration & Sharing
- Share evaluation results with team members or at public events
- Enable stakeholders to review metrics without access to the IT framework
- Create standardized reports for management
- Facilitate a review of model performance

Analysis & Visualization
- Import data into BI tools
- Create dashboards for performance tracking
- Perform complex SQL analytics in BigQuery
- Enable historical trend analysis

Business & Technical Value:

Operational Benefits:
- Track model performance over time
- Compare different model versions
- Identify degradation patterns early

Research & Development:
- Build training datasets for future models
- Create benchmarks for model selection
- Document experimental results
- Support academic publications

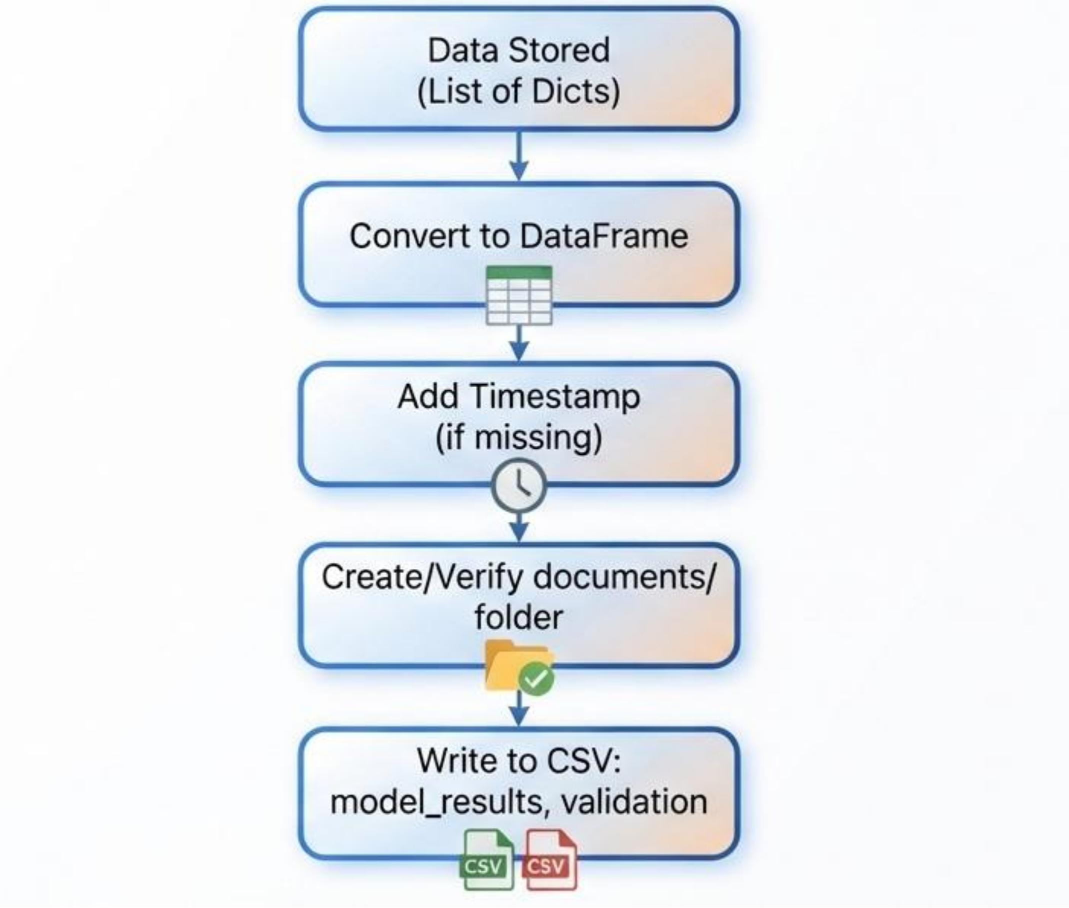
DataFrame – The Heart of Data Processing

Definition: DataFrame is two-dimensional, labeled data structure in Python
Purpose: Organizes evaluation results in a structured format
Origin: Pandas library (Python's data manipulation toolkit)

Use DataFrames for:

- Structured Storage
  - Columns represent the metrics
  - Rows represent individual evaluations
- Built-in Operations
  - Easy filtering and sorting
  - Aggregation functions (mean, sum, count)
  - Quick conversion to CSV or JSON
- BigQuery Compatibility
  - Direct conversion to BigQuery table schema
  - Supports various data types

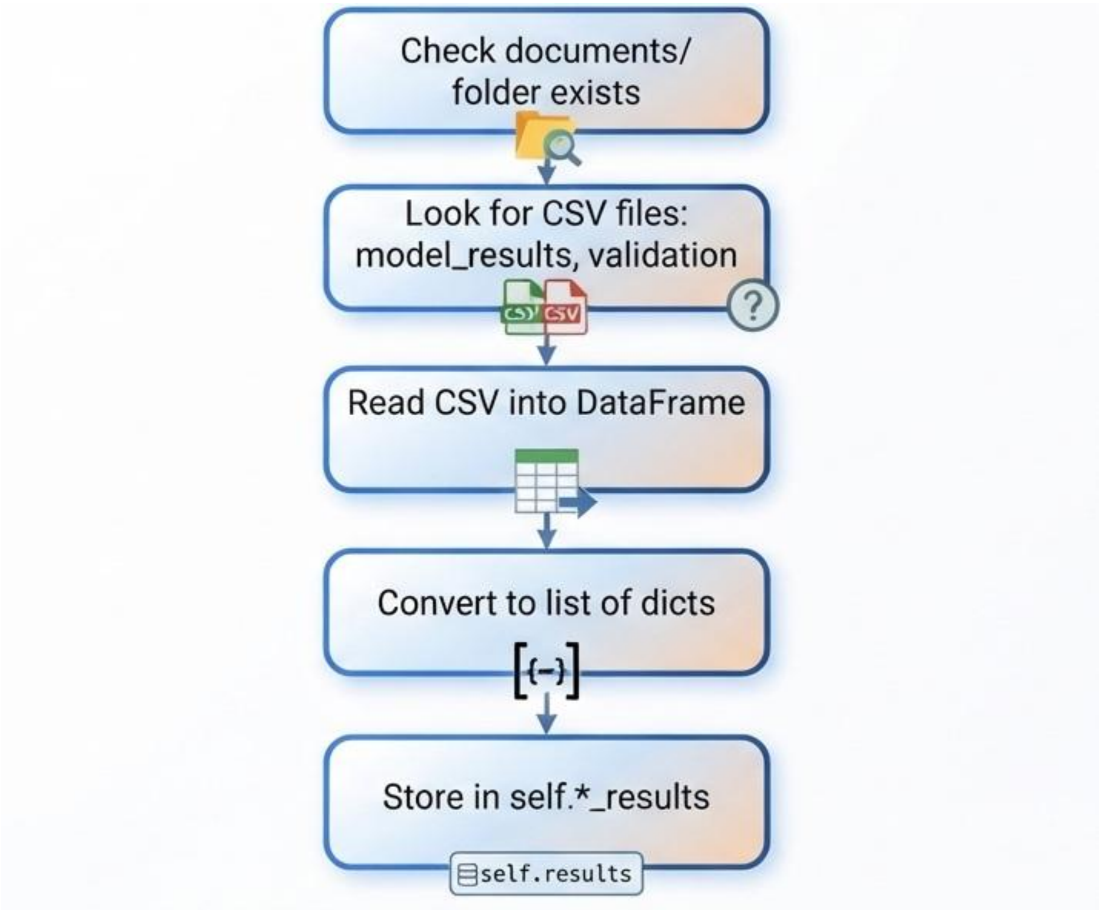
Data CSV Export Flow



Key Features:

- Automatic folder creation
- Timestamp tracking for versioning
- Create Two files:
  - model_results.csv
  - validation_results.csv
- Human-readable format for inspection

Data CSV Import Flow



Key Features:

- Automatic file detection
- Handling of missing files
- Integration with existing data structures
- No data loss on import

Error Handling:

- File not found
  - informative message, continue
- Empty CSV
  - import nothing, continue
- Corrupted file
  - exception handling, continue

Introduction to Google Cloud Platform (GCP)

Google Cloud Platform (GCP) is a suite of cloud computing services provided by Google, offering infrastructure, platform, and software solutions. It enables organizations to build, deploy, and scale applications without managing physical hardware.

Service	Purpose	Relevance
BigQuery	Serverless data warehouse for analytics	Stores and queries LLM evaluation results
IAM	Identity and Access Management	Controls permissions for service accounts
Cloud Storage	Object storage for unstructured data	Optional future use for model artifacts
Cloud SDK	Command-line tools for managing resources	Enables automation and scripting

Why Google Cloud?

1. Serverless Architecture
- No infrastructure management required
- Automatic scaling for growing datasets
- Pay only for resources actually used

2. Enterprise-Grade Security
- TLS encryption for all data in transit
- IAM for granular access control
- Compliance with global standards

3. Data Analytics Power
- BigQuery processes petabyte-scale datasets
- Real-time and batch query support
- SQL-based analytics for evaluation data

4. Integration with Python
- Native Python client libraries
- pandas-gbq for seamless DataFrame integration
- gcloud CLI for automation

Google BigQuery - Cloud Data Warehouse

Definition: Google Cloud BigQuery is serverless, highly-scalable data warehouse
Purpose: Store and query large datasets without infrastructure management
Key Feature: SQL-compatible, petabyte-scale analysis

Why We Use BigQuery:

1. Scalability
- No storage limits
- Handles millions of rows effortlessly
- Automatic partitioning for performance

2. Accessibility
- Accessible from anywhere with credentials
- Team collaboration without file sharing
- Integration with Python via pandas-gbq

3. Cost-Effective
- Pay only for storage and queries used
- No server maintenance costs
- Free tier available for small datasets

Identity and Access Management (IAM) & Service Account Authentication

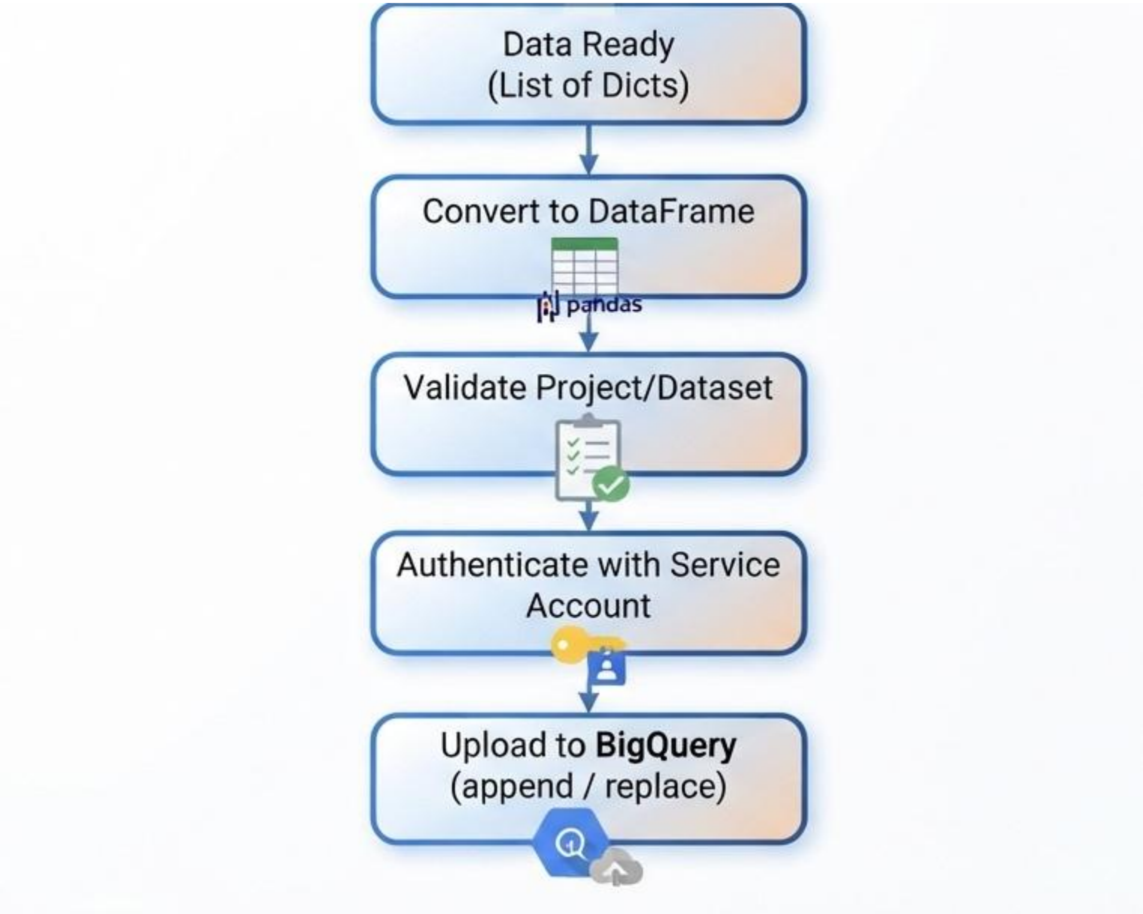
Definition: Google Cloud's identity and access management system
Purpose: Controls who can access GCP resources
Key Concept: Least privilege principle

Service Account Role:
- A special type of Google account for applications
- Used by Python script to authenticate with BigQuery
- Provides secure, programmatic access

How Authentication Works:
- Service account key (JSON) stored in "credentials" folder
- Python code loads credentials from environment variable
- Each request to BigQuery includes the authentication token
- IAM verifies permissions before allowing access

Key Management:
- Key files kept secure (never commit to Git)
- Environment variable for path: GOOGLE_APPLICATION_CREDENTIALS
- Automatic token refresh for long-running sessions

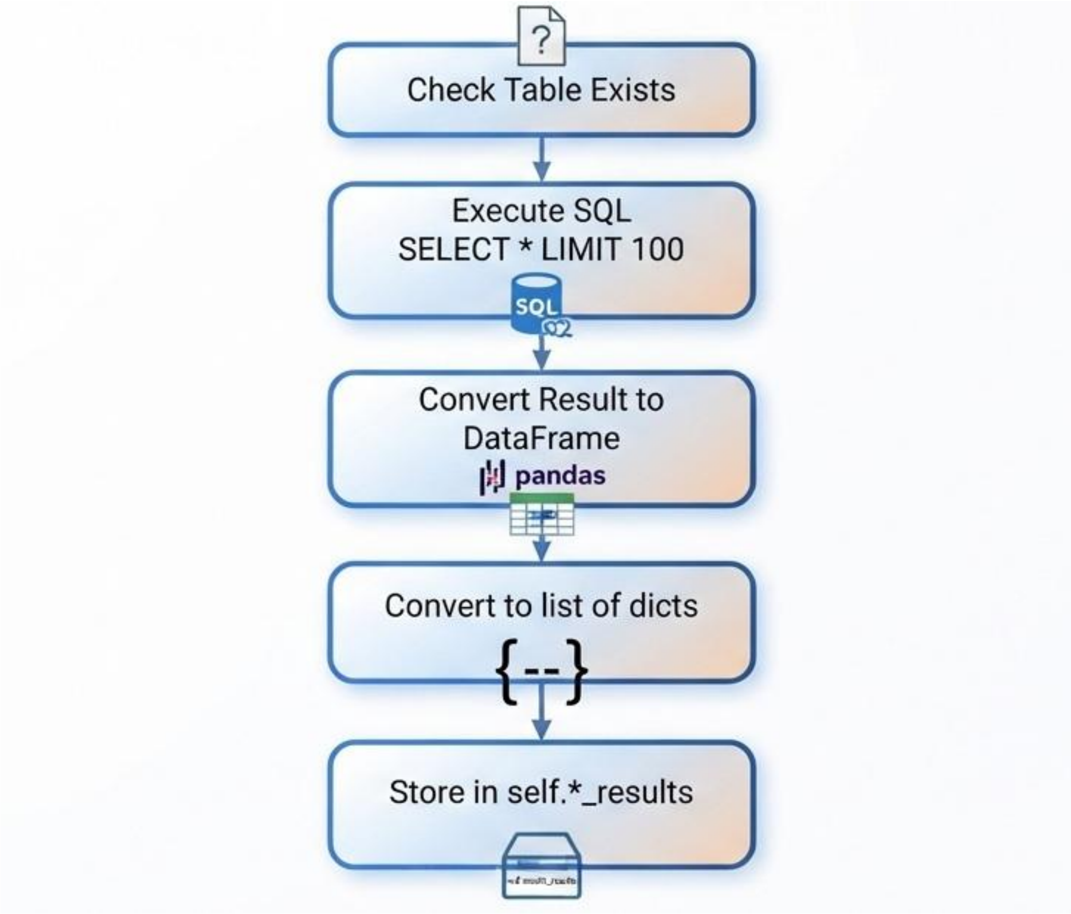
BigQuery Export Flow



Key Features:
- Schema-aware upload
- Progress bar for large datasets
- Multiple tables supported
- Configurable insert behavior

Mode	Behavior
append	Adds new rows, preserves existing
replace	Drops table, recreates with new data
fail	Errors if table exists

BigQuery Import Flow



Key Features:
- Table existence check before import
- SQL query with LIMIT (configurable)
- Graceful handling of missing tables
- Automatic schema detection

Table	Contains
model_results	Evaluation metrics from AI model testing
validation_results	Human validation results with labels

CSV vs BigQuery Flow Comparison

Aspect	CSV Export	BigQuery Export
Storage	Local file system	Google Cloud
Access	Any text editor	SQL queries
Collaboration	File sharing	Shared database
Scalability	Limited by disk	Petabyte-scale
Query Power	Basic (pandas)	SQL (complex)
Speed	Very fast	Variable (serverless)

LIVE DEMO

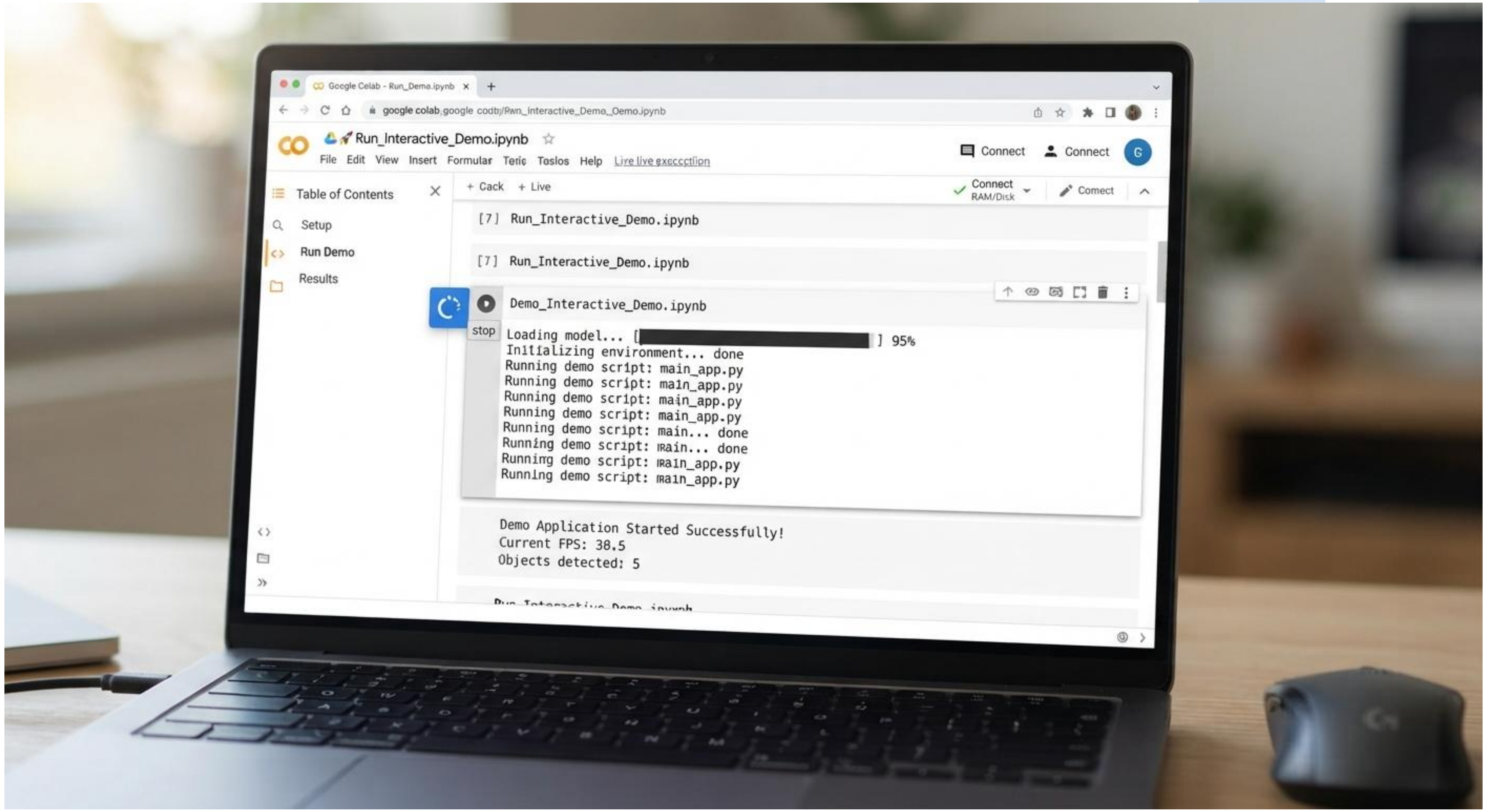

Google Colab - Run_Demo.ipynb
google colab.google codb/Rwn_interactive_Demo_Oemo.ipynb
Run_Interactive_Demo.ipynb
File Edit View Insert Formular Teric Toslos Help Lire live exeeution
Connect
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Table of Contents
Setup
Run Demo
Results
[7] Run_Interactive_Demo.ipynb
[7] Run_Interactive_Demo.ipynb
Demo_Interactive_Demo.ipynb
stop
Loading model... [ ] 95%
Initializing environment... done
Running demo script: main_app.py
Running demo script: main_app.py
Running demo script: main_app.py
Running demo script: main_app.py
Running demo script: main... done
Running demo script: main... done
Running demo script: main_app.py
Running demo script: main_app.py
Demo Application Started Successfully!
Current FPS: 30.5
Objects detected: 5

References

- DataFrame – https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.html
- Pandas CSV export – https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.to_csv.html
- Pandas CSV import – https://pandas.pydata.org/docs/reference/api/pandas.read_csv.html
- Google Cloud Documentation – https://docs.cloud.google.com/docs
- Google BigQuery Documentation – https://docs.cloud.google.com/bigquery/docs
- Identity and Access Management documentation – https://docs.cloud.google.com/iam/docs
- Pandas BigQuery Dataset export – https://pandas-gbq.readthedocs.io/en/latest/api.html#pandas_gbq.to_gbq
- Pandas BigQuery Dataset import – https://pandas-gbq.readthedocs.io/en/latest/api.html#pandas_gbq.read_gbq
- Google BigQuery export table data to Cloud storage – https://docs.cloud.google.com/bigquery/docs/exporting-data#python

Thank you

Mariyan Vasilev Apostolov

mariyan.apostolov89@gmail.com